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Sub - Research Methodology (CS601)

1) A researcher wants to test whether the average regular gasoline price in the midwest region is higher than \$ 2.45

Given — sample size $n = 25$
sample mean $\bar{x} = 2.65$
sample standard deviation $s = 0.35$
significance $\alpha = 0.01$ (99% confidence level)
critical value (t-table) = 2.49

a) Null and Alternative Hypothesis

→ since the researcher wants to test if the mean is higher, this is a right-tailed test.

$$H_0 : \mu = 2.45$$
$$H_1 : \mu > 2.45$$

where,
μ = population mean gasoline price

b) Test hypothesis

→ since population standard deviation is not given and sample size is small ($n < 30$), we use t-test.

(P.T.O)

Step 1:- Test statistic formula

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$t = \frac{2.65 - 2.45}{0.35/\sqrt{25}}$$

$$t = \frac{0.20}{0.35/5}$$

$$t = \frac{0.20}{0.07}$$

$$t = 2.857$$

Step-2 - Decision Rule

- calculated value = 2.857
- critical value (given) = 2.49

since, $2.857 > 2.49$

- conclusion:- we reject the null hypothesis at 1% significance level.

There is sufficient evidence to conclude that the average gasoline price in the mid west region is higher than \$ 2.45.

2> Benefits of Literature in R.M

→ Literature review plays and important role in research its main benefits are:-

i> Provides Background Knowledge :-
Helps the Researcher understand previous work related to topic.

ii> Identifies Reesearch Gaps :-
Shows what has not been studied or needs improvement.

iii> Avoids Duplication :- prevents repeating the same research unnecessarily

iv> Helps in Formulating Hypothesis :-
Guides in developing research questions and hypotheses.

v> Improves Research Design :-
suggests suitable methods, tools, and techniques.

Thus, literature review strengthens the foundation and quality of research.